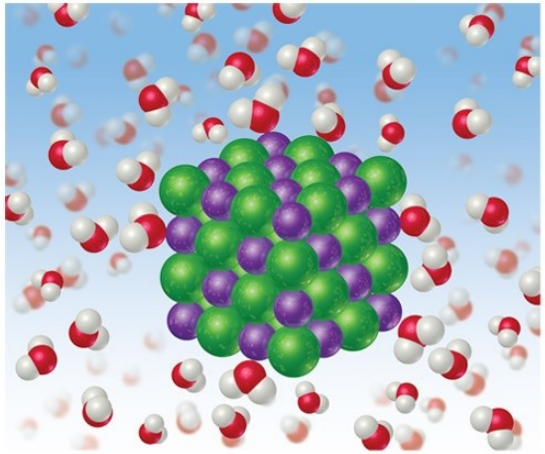
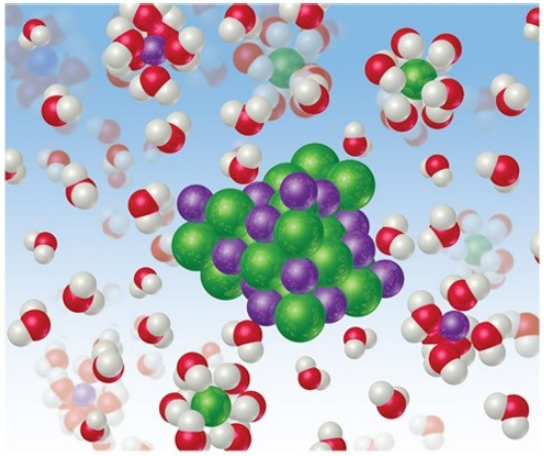
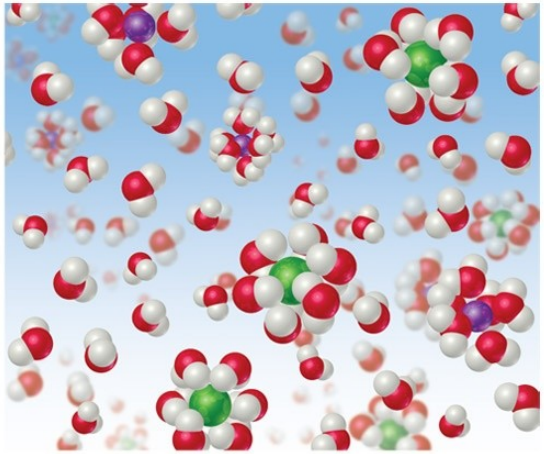
Electrolytes

- **Solutes which improve the conductivity of a solution are called electrolytes.**
- **A solution which conducts electricity is known as an electrolyte solution.**
- **Pure water is a non-electrolyte. Water containing salt such as NaCl is an electrolyte solution (due to the presence of ions in solution)**

Electrolytes

- **NaCl and other salts don't just dissolve, they dissociate in water.**
- **Ions that were packed together in the crystal separate upon dissolving and become individual (solvated) ions in the solution.**

Electrolytes

- **We write this in form of dissociation equations (to keep track of the number of ions per formula unit)**

$$NaCl(s) \leftrightarrow Na^{+}(aq) + Cl^{-}(aq)$$

$$K_2CO_3(s) \leftrightarrow 2\ K^{+}(aq) + CO_3^{2-}(aq)$$

- **(s) denotes solid; (aq) denotes aqueous (in water)**

Ion Concentrations

- A solution of just NaCl and water has equal concentrations of sodium and chloride ions
- For example in a 0.1 M NaCl we have:

$$NaCl(s) \leftrightarrow Na^{+}(aq) + Cl^{-}(aq)$$
$$0.1\ mol \leftrightarrow 0.1\ mol + 0.1\ mol$$

- Thus in a 0.1 M NaCl the concentration of chloride ions is 0.1 mol/L and the concentration of sodium ions is also 0.1 mol/L.

Ion Concentrations

- How about a 0.1 M $CaCl_2$ we have:

$$CaCl_2(s) \leftrightarrow Ca^{2+}(aq) + 2Cl^-(aq)$$

$$0.1\ mol \leftrightarrow 0.1\ mol + 0.2\ mol$$

- Thus in a 0.1 M $CaCl_2$ solution concentration of chloride ions is 0.2 mol/L and the concentration of sodium ions is 0.1 mol/L.

Exercise

- **What has the greatest amount of chloride ions**

a) 1.2 L of 0.50 M NaCl

b) 1.1 L of 0.40 M $CaCl_2$

c) 0.75 L of 0.40 M $AlCl_3$

d) 2.0 L of 0.50 M $KClO_3$

e) 30 L of 0.001 M Cl_2

Exercise

- **What has the greatest amount of chloride ions**

a) 1.2 L of 0.50 M NaCl

b) 1.1 L of 0.40 M $CaCl_2$

c) 0.75 L of 0.40 M $AlCl_3$

d) 2.0 L of 0.50 M $KClO_3 \leftrightarrow K^+(aq) + ClO_3^-(aq)$

e) 30 L of 0.001 M Cl_2

$M = \frac{n}{V}$ n(chloride) = M(chloride)x V

mol

Ionization

- Sugars dissolve in water but do not dissociate (don't form ions) Why? The bonds in the sugar molecules are **covalent**, not ionic!

$C_6H_{12}O_6(s) \rightarrow C_6H_{12}O_6(aq)$

glucose non-electrtolye

- However if the bonds are **very polar** they might break in water leading to ions after all:

$HCl(g) \rightarrow H^+(aq) + Cl^-(aq)$

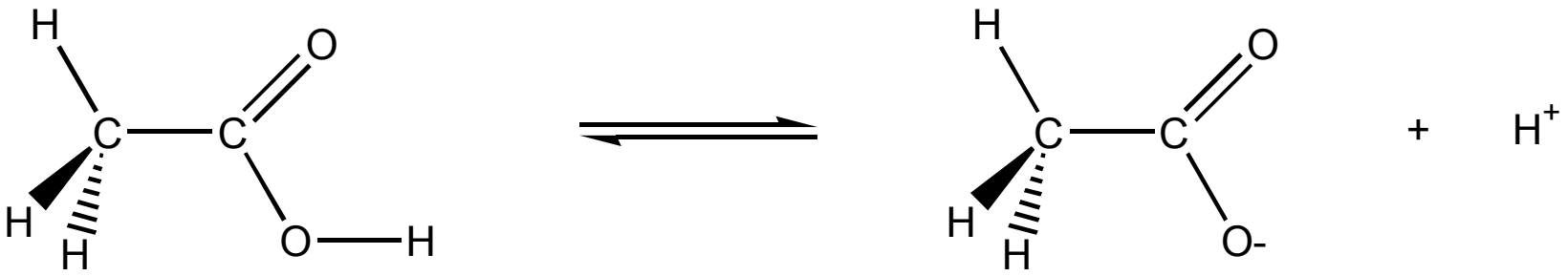
Hydrogen chloride hydrochloric acid

Ionization

- The HCl molecules have been split apart by the water resulting in the formation of ions. We call this process **ionizing in water.**
- Acids are electrolytes! **Strong acids ionize completely and their solutions contain high concentrations of H^+ ions.**
- Strong acids are strong electrolytes.
- **For example HNO_3, H_2SO_4, $HClO_4$**

Ionization

- Weak acids ionize incompletely and their solutions contain lower concentrations of H^+ ions.
- Weak acids are weak electrolytes.



So what are acids anyway?

Arrhenius Theory

Acids are proton donors whilst bases are hydroxide ion donors.

Brønsted Theory

Acids are proton donors whilst bases are proton acceptors.

Lewis Theory

Acids are electron pair acceptors whilst bases are electron pair donors.

The pH Scale

- The pH scale is a convenient way of referring to the degree of acidity of a solution.
- **pH is the negative decadic logarithm of the H^+ ion concentration: $pH = -\log [H^+]$**

The pH Scale

- So pure water has a pH of 7 meaning the concentration of **H^+** is **10^{-7} mol/l = 0.0000001 M**
- **This comes from the ionization of water itself (very slight)**

$$H_2O(l) \rightleftarrows H^+(aq) + OH^-(aq)$$

- **We see from this equation that there is an equal concentration of** H^+ and OH^- (neutral)

Chemical equilibrium

- For the general reaction **aA + bB + …⇆ eE + gG**

$$\frac{[\boldsymbol{E}]^{\mathrm{e}}[\boldsymbol{G}]^{\mathrm{g}}}{[\boldsymbol{A}]^{\mathrm{a}}[\boldsymbol{B}]^{\mathrm{b}}} = \mathbf{k_{eq}}$$

- The fraction $\frac{[E]^{\mathrm{e}}[G]^{\mathrm{g}}}{[A]^{\mathrm{a}}[B]^{\mathrm{b}}}$ is the equilibrium constant expression and its numerical value k_{eq} is the equili constant. Note that coefficients in the reaction equ become exponents in the equilibrium constant expression.

Chemical equilibrium

- The fraction $\frac{[E]^{\mathrm{e}}[G]^{\mathrm{g}}}{[A]^{\mathrm{a}}[B]^{\mathrm{b}}}$ is the equilibrium constant expression.
- The square brackets indicate that we agree on usi the numerical value for the molar concentrations o reactants and products if in solution and the nume value for partial pressure (in atm) for gaseous reac and/or products. For pure liquids or pure solids we enter unity

Equilibrium

$$H_2O(l) \leftrightarrows H^+(aq) + OH^-(aq)$$

$$\frac{[H^+]^1[OH^-]^1}{[H_2O]^1} = k$$

Because so little water ionizes, the concentration of water remains effectively constant at 55.5 mol/L

$$[H^+]\,[OH^-] = [H_2O]k = K_w = 1.0 \times 10^{-14}$$

Ion-Product Constant

$$K_w = [H_3O^+]\,[OH^-] = [H^+]\,[OH^-] = 1.0 \times 10^{-14}$$

- This special equilibrium constant is referred to as the ion-product constant for water, K_w(25°C)

- This is true for all aqueous solutions, not just pure water! **This means if $[H^+]$ goes up (addition of acid), $[OH^-]$ must decrease to maintain $K_w = 1.0 \times 10^{-14}$ and vice versa.**

Ion-Product Constant

- $K_w = 1.0 \times 10^{-14} = [H_3O^+]\,[OH^-]$ **(1)**
- **Let's take the –log of expression (1).**
- $-\log K_w = -\log(1.0 \times 10^{-14}) = -\log\{[H_3O^+][OH^-]\}$
- $pK_w = 14 = -\{\log[H_3O^+] + \log[OH^-]\}$
- $pK_w = 14 = pH + pOH$ **(in all aqueous solutions)**

pH and pOH

These are the pH values for several common substances.

	$[H^+]$ (*M*)	pH	pOH	$[OH^-]$ (*M*)
	$1\ (1\times10^{-0})$	**0.0**	**14.0**	1×10^{-14}
Gastric juice	1×10^{-1}	**1.0**	**13.0**	1×10^{-13}
Lemon juice	1×10^{-2}	**2.0**	**12.0**	1×10^{-12}
Cola, vinegar	1×10^{-3}	**3.0**	**11.0**	1×10^{-11}
Wine Tomatoes	1×10^{-4}	**4.0**	**10.0**	1×10^{-10}
Banana Black coffee	1×10^{-5}	**5.0**	**9.0**	1×10^{-9}
Rain Saliva	1×10^{-6}	**6.0**	**8.0**	1×10^{-8}
Milk Human blood, tears	1×10^{-7}	**7.0**	**7.0**	1×10^{-7}
Egg white, seawater	1×10^{-8}	**8.0**	**6.0**	1×10^{-6}
Baking soda	1×10^{-9}	**9.0**	**5.0**	1×10^{-5}
Borax	1×10^{-10}	**10.0**	**4.0**	1×10^{-4}
Milk of magnesia Lime water	1×10^{-11}	**11.0**	**3.0**	1×10^{-3}
Household ammonia	1×10^{-12}	**12.0**	**2.0**	1×10^{-2}
Household bleach	1×10^{-13}	**13.0**	**1.0**	1×10^{-1}
NaOH, 0.1 *M*	1×10^{-14}	**14.0**	**0.0**	$1\ (1\times10^{-0})$

Figure 14.4

Relating $[H_3O^+]$, $[OH^-]$, pH and pOH

- **Example**: the pH of rain water was measured to be 4.35 whilst that of aqueous ammonia was found to be 11.28
- Determine $[H_3O^+]$ in water and $[OH^-]$ in ammonia,

Relating $[H_3O^+]$, $[OH^-]$, pH and pOH

- **Example**: the pH of rain water was measured to be 4.35 whilst that of aqueous ammonia was found to be 11.28
- Determine $[H_3O^+]$ in water and $[OH^-]$ in ammonia,
- $4.35 = pH = -\log [H_3O^+] = -\log [H^+]$
- $-4.35 = \log [H^+]$
- $10^{-4.35} = [H^+] = 4.5 \times 10^{-5}$ mol/L

Relating $[H_3O^+]$, $[OH^-]$, pH and pOH

- **Example**: the pH of rain water was measured to be 4.35 whilst that of aqueous ammonia was found to be 11.28
- Determine $[H_3O^+]$ in water and $[OH^-]$ in ammonia, pH + pOH = 14, pOH = 2.72
- $2.72 = pOH = -\log [OH^-]$
- $-2.72 = \log [OH^-]$
- $10^{-2.72} = [OH^-] = 1.9 \times 10^{-3}$ mol/L

Therefore we can easily calculate the pH of aqueous solutions of strong mono protic acids such as HCl, HBr, HI, HNO_3, $HClO_4$

Example: Calculate the pH of a 0.0025 M HCl

What is the pH of a saturated solution of Ca(OH)2
Solubility is 0.16 g in 100 ml solution at RT

Class Exercise

Calculate the pH of a 0.0025 M HCl

$HCl(g) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$

All of the HCl ionizes (strong acid) and at equilibrium $H_3O^+(aq) = 0.0025$ M

And pH = -log(0.0025) = 2.6

Class Exercise

What is the pH of a saturated solution of $Ca(OH)_2$
Solubility is 0.16 g in 100 ml solution at RT

Class Exercise

What is the pH of a saturated solution of $Ca(OH)_2$
Solubility is 0.16 g in 100 ml solution at RT

Strategy: We need molarity of hydroxide ions in solution to calculate pOH and ultimately pH. So lets convert the solubility of $Ca(OH)_2$ into mol/L and then relate the molarity of $Ca(OH)_2$ to the molarity of hydroxide.

Class Exercise

What is the pH of a saturated solution of $Ca(OH)_2$
Solubility is 0.16 g in 100 ml solution at RT

$$0.16\ \frac{g}{100\ mL} = 0.16\ \frac{10\ g}{1000\ mL} = 1.6\ g/L\ Ca(OH)_2(aq)$$

$$1.6\ \frac{g}{L}\ \times \frac{mol}{74.1\ g} = 0.0216\ \frac{mol}{L}\ Ca(OH)_2$$ $Ca(OH)_2(s) \leftrightarrow Ca^{2+}(aq) + 2OH^-(aq)$

$$0.0216\ \frac{mol}{L}\ Ca(OH)_2(aq) \times \frac{2\ mol\ OH^-}{1\ mol\ Ca(OH)_2} = 0.0432\ \frac{mol}{L}\ OH^-(aq)$$

$pOH = -\log[OH^-] = -\log[0.0432] = 1.36$

And $pH = 14 - pOH = 12.6$

What about the pH of weak acids like acetic acid or HF and what about acids that can give off more than one hydrogen ion upon ionization H_3PO_4, H_2SO_4

Dissociation Constants

- For a generalized acid dissociation,

$$HA(aq) + H_2O(l) \rightleftharpoons A^-(aq) + H_3O^+(aq)$$

the equilibrium expression would be

$$K = \frac{[H_3O^+]\,[A^-]}{[HA][H_2O]}$$

For weak acids the concentration of water remains essentially constant at 55.5 M because so few water molecules react with the acid.

Dissociation Constants

- For a generalized acid dissociation,

$$HA(aq) + H_2O(l) \rightleftharpoons A^-(aq) + H_3O^+(aq)$$

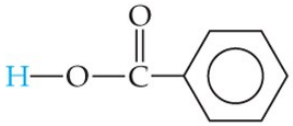
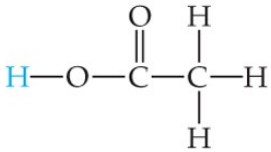
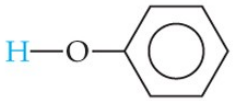
the equilibrium expression would be

$$K_a = \frac{[H_3O^+]\,[A^-]}{[HA]}$$

We can combine , $K[H_2O]$ into a new constant K_a

Dissociation Constants

The greater the value of K_a, the stronger the acid.

Acid	Structural Formula	Conjugate Base	Equilibrium Reaction	K_a
Hydrofluoric (HF)	H—F	F^-	$HF(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + F^-(aq)$	6.8×10^{-4}
Nitrous (HNO_2)	H—O—N=O	NO_2^-	$HNO_2(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + NO_2^-(aq)$	4.5×10^{-4}
Benzoic ($HC_7H_5O_2$)		$C_7H_5O_2^-$	$HC_7H_5O_2(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + C_7H_5O_2^-(aq)$	6.3×10^{-5}
Acetic ($HC_2H_3O_2$)		$C_2H_3O_2^-$	$HC_2H_3O_2(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + C_2H_3O_2^-(aq)$	1.8×10^{-5}
Hypochlorous (HClO)	H—O—Cl	ClO^-	$HClO(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + ClO^-(aq)$	3.0×10^{-8}
Hydrocyanic (HCN)	H—C≡N	CN^-	$HCN(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CN^-(aq)$	4.9×10^{-10}
Phenol (HC_6H_5O)		$C_6H_5O^-$	$HC_6H_5O(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + C_6H_5O^-(aq)$	1.3×10^{-10}

*The proton that ionises is shown in blue.

pH of weak acids

Example: What is the pH of a 0.100 M acetic acid solution:

$$CH_3COOH\ (l) + H_2O\ (l) \rightleftharpoons H_3O^+(aq) + CH_3COO^-(aq)$$

We also know the acid dissociation constant

$$k_a = 1.8 \times 10^{-5} = \frac{[H_3O^+][CH_3COO^-]}{[CH_3COOH]}$$

pH of weak acids

Example: What is the pH of a 0.100 M acetic acid solution:

$$CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$$

Initial conc.:	**0.1000M**	**0**	**0**

pH of weak acids

Example: What is the pH of a 0.100 M acetic acid solution:

$$CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$$

Initial conc.:	**0.1000M**		**0**	**0**
Change:	**-x M**		**+x M**	**+ x M**

pH of weak acids

Example: What is the pH of a 0.100 M acetic acid solution:

$$CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$$

Initial conc.:	**0.1000M**	**0**	**0**
Change:	**-x M**	**+x M**	**+ x M**
Equilibrium:	**(0.100 – x)M**	**x M**	**x M**

Now relate the equilibrium concentrations to the acid dissociation constant.

pH of weak acids

Example: What is the pH of a 0.100 M acetic acid solution:

	$CH_3COOH + H_2O$	$\rightleftharpoons$	H_3O^+	$+ CH_3COO^-$
Initial conc.:	**0.1000M**		**0**	**0**
Change:	**-x M**		**+x M**	**+ x M**
Equilibrium:	**(0.100 – x)M**		**x M**	**x M**

$$k_a = 1.8 \times 10^{-5} = \frac{[CH_3COO^-][H_3O^+]}{[CH_3COOH]} = \frac{x^2}{0.100 - x}$$

pH of weak acids

$k_a = 1.8 \times 10^{-5} = \frac{[CH_3COO^-][H_3O^+]}{[CH_3COOH]} = \frac{x^2}{0.100-x} \approx \frac{x^2}{0.100}$ if x<< 0.1

$1.8 \times 10^{-6} = x^2$ and $x = \sqrt[2]{1.8\ x\ 10^{-6}}$

x = 0.00134 which is 2 orders of magnitude smaller than 0.1

x<< 0.1 (simplifying assumption holds)

Last we relate x to pH, in this case x = $[H^+]$ and pH = -log[x]

pH = -log[0.00134] = 2.9

Not that if our simplifying assumption fails we do need to solve the quadratic equation!

pH of a weak base

Example: What is the pH of 0.150 M ammonia?

	$NH_3 + H_2O$	$\rightleftharpoons$	NH_4^+	$+ OH^-$
Initial conc.:	**0.150M**		**0**	**0**
Change:	**-x M**		**+x M**	**+ x M**
Equilibrium:	**(0.150 – x)M**		**x M**	**x M**

$$k_b = 1.8 \times 10^{-5} = \frac{[NH_4^+][OH^-]}{[NH_3]} = \frac{x^2}{0.150 - x}$$

Note that the numerical value for k_b being the same as k_a for acetic acid is a coincidence.

pH of a weak base

$k_b = 1.8 \times 10^{-5} = \frac{[NH_4^+][OH^-]}{[NH_3]} = \frac{x^2}{0.150 - x} \approx \frac{x^2}{0.150}$ if x<< 0.15

$2.7 \times 10^{-6} = x^2$ and $x = \sqrt[2]{2.7\ x\ 10^{-6}}$

x = 0.00164 which is 2 orders of magnitude smaller than 0.15

x<< 0.15 (simplifying assumption holds)

Last we relate x to pOH, in this case x = $[OH^-]$ and pOH = -log[x]

pOH = -log[0.00164] = 2.8 and pH = 14 - 2.8 = 11.2

The pH of polyprotic acids

Sulfuric acid-a special case

What is the pH of 0.50 M H_2SO_4 K_{a1} >>1, k_{a2} = 1.1x10^{-2}

$$\mathbf{H_2SO_4(l) + H_2O(l) \rightarrow HSO_4^-(aq) + H_3O^+(aq)}\ K_{a1} >> 1$$

What is the pH of 0.50 M H_2SO_4 K_{a1} >>1, k_{a2} = 1.1x10^{-2}

	$H_2SO_4(l) + H_2O(l)$ →	$HSO_4^-(aq)$ +	$H_3O^+(aq)$ K_{a1} >>1
Initial conc.:	**0.50 M**	**0**	**0**
Change:	**-0.50 M**	**+0.50 M**	**+0.50 M**
After 1th ionization:	**~ 0 M**	**0.50 M**	**0.50 M**

$$HSO_4^-(aq) + H_2O(l) \rightarrow SO_4^{2-}(aq) + H_3O^+(aq) \quad k_{a2} = 1.1\text{x}10^{-2}$$

What is the pH of 0.50 M H_2SO_4 K_{a1} >>1, k_{a2} = 1.1x10^{-2}

	$H_2SO_4(l) + H_2O(l)$ →	$HSO_4^-(aq)$ +	$H_3O^+(aq)$ K_{a1} >>1
Initial conc.:	**0.50 M**	**0**	**0**
Change:	**-0.50 M**	**+0.50 M**	**+0.50 M**
After 1th ionization:	**~ 0 M**	**0.50 M**	**0.50 M**

	$HSO_4^-(aq) + H_2O(l)$ →	$SO_4^{2-}(aq)$ +	$H_3O^+(aq)$ k_{a2} = 1.1x10^{-2}
After 1th ionization	**0.50 M**	**0**	**0.50 M**
Change:	**-x M**	**+x M**	**+x M**
After 2 nd ionization: (equilibrium)	**(0.50 –x) M**	**+x M**	**(0.50 +x) M**

$$1.1 \times 10^{-2} = \frac{[SO_4{}^{2-}][H_3O^+]}{[HSO_4{}^-]} = \frac{x(0.50+x)}{0.50-x} \approx x \text{ if } x << 0.50$$

x = 1.1 x 10^{-2} which is about 2 % of 0.5 which is just o.k.

$[H^+]$ = (0.5 + x) = 0.511 and pH = -log 0.511 = 0.29

The pH of salt solutions

$NaCl(s) \leftrightarrow Na^+(aq) + Cl^-(aq)$ pH = 7

Not surprising since NaCl is the product of an acid base neutralization:

$NaOH(aq) + HCl(aq) \leftrightarrow NaCl(aq) + H_2O(l)$

Strong base + strong acid→ neutral salt

But…

The pH of salt solutions

$NaCl(s) \leftrightarrow Na^+(aq) + Cl^-(aq)$ pH = 7

$CH_3COONa(s) \leftrightarrow Na^+(aq) + CH_3COO^-(aq)$ pH > 7

$NH_4Cl(s) \leftrightarrow NH_4^+(aq) + Cl^-(aq)$ pH < 7

$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$

Strong base + strong acid→ neutral salt

$NaOH(aq) + CH_3COOH\ (aq) \rightarrow CH_3COONa\ (aq)$

Strong base + weak acid → basic salt

$NH_3(aq) + HCl(aq) \rightarrow NH_4Cl\ (aq)$

Weak base + strong acid → acidic salt

The pH of 0.100 M $NH_4Cl(aq)$

1.) dissociate the salt in water

$NH_4Cl(s) \leftrightarrow NH_4^+(aq) + Cl^-(aq)$ pH < 7

2.) hydrolyze the ion of the weak base or acid (react with water) and set up ICE table

$$NH_4^+(aq) + H_2O(l) = NH_3(aq) + H_3O^+(aq)$$

Initial conc.:	**0.100 M**	**0**	**0**
Change:	**-x M**	**+x M**	**+ x M**
Equilibrium:	**(0.100 – x)M**	**x M**	**x M**

Need ka for $NH_4^+(aq)$ = $\frac{[NH_3][H_3O^+]}{[NH_4^+]}$ **but only given kb for $NH_3(aq)$** $1.8 \times 10^{-5} = \frac{[NH_4^+][OH^-]}{[NH_3]}$

$$\text{Ka } (NH_4^+) = \frac{[NH_3][H_3O^+]}{[NH_4^+]} = \frac{[NH_3][H_3O^+][OH^-]}{[NH_4^+][OH^-]} = \frac{[NH_3]kw}{[NH_4^+][OH^-]} = \frac{kw}{kb\ (NH_3)}$$

The pH of 0.100 M $NH_4Cl(aq)$

$$NH_4^+(aq) + H_2O(l) = NH_3(aq) + H_3O^+(aq)$$

Initial conc.:	**0.100 M**	**0**	**0**
Change:	**-x M**	**+x M**	**+ x M**
Equilibrium:	**(0.100 – x)M**	**x M**	**x M**

More general: $K_a \times k_b = k_w$ for conjugated acid base pairs

so $Ka(NH_4^+) = \frac{kw}{kb\ (NH_3)} = \frac{10^{-14}}{1.8\ x\ 10^{-5}} = 5.5 \times 10^{-10}$

And ka(NH_4^+) = $5.5 \times 10^{-10} = \frac{[NH_3][H_3O^+]}{[NH_4^+]} = \frac{x^2}{0.100 - x} \approx \frac{x^2}{0.100}$ if $x << 0.1$

$x = \sqrt{5.5\ x\ 10^{-11}} = 7.4 \times 10^{-6} << 0.1$

$x = [H^+]$ and pH = -log[x] pH = -log[7.45×10^{-6}] = 5.1 (mildly acidic)

Buffers

Assume we prepared a solution in which the molarity of both acetic acid and acetate ions at equilibrium was 1.00 M.

a) How would we prepare such a solution?

b) What would be the pH of such a solution?

K_a(acetic acid) = 1.8×10^{-5}

$CH_3COO^-(aq)$	$CH_3COOH(aq)$

Buffers

a) Mix acetic acid with sodium acetate, or titrate 2M acetic acid with NaOH until $[CH_3COO^-] = [CH_3COOH] = 1.00$ M

b) $CH_3COOH(aq) + H_2O(l) = CH_3COO^-(aq) + H_3O^+(aq)$

$$K_a = 1.8 \times 10^{-5} = \frac{[CH_3COO^-][H_3O^+]}{[CH_3COOH]}$$

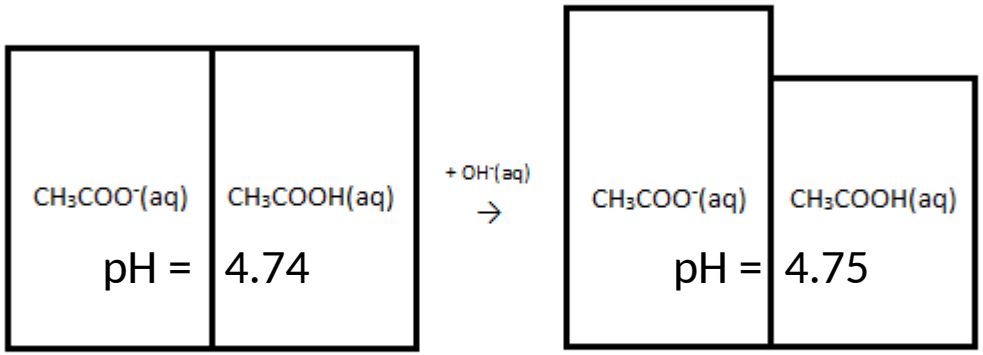
and if $[CH_3COO^-] = [CH_3COOH]$

$K_a = 1.8 \times 10^{-5} = [H_3O^+]$

And $pH = -\log[1.8 \times 10^{-5}] = 4.74$

Buffers

- What happens to such a system if we add a small amount of strong base (e.g. add 0.01 mol NaOH)



$CH_3COOH(aq) + OH^-(aq) = CH_3COO^-(aq) + H_2O(l)$

$K_a = 1.8 \times 10^{-5} = \frac{[CH_3COO^-][H_3O^+]}{[CH_3COOH]} = \frac{[1.01][H_3O^+]}{[0.99]}$

and $H_3O^+(aq) = 1.8 \times 10^{-5} \frac{[0.99]}{[1.01]}$ and pH = 4.75 (note without buffer pH = 12!!!)

Buffers

- What if we added a small amount of strong acid (e.g. add 0.01 mol HCl)

$CH_3COO^-(aq)$	$CH_3COOH(aq)$
pH =	4.73

$\xleftarrow{+\,H_3O^+(aq)}$

$CH_3COO^-(aq)$	$CH_3COOH(aq)$
pH =	4.74

$CH_3COO^-(aq) + H_3O^+(aq) = CH_3COOH(aq) + H_2O(l)$

$$K_a = 1.8 \times 10^{-5} = \frac{[CH_3COO^-][H_3O^+]}{[CH_3COOH]} = \frac{[0.99][H_3O^+]}{[1.01]}$$

and $H_3O^+(aq) = 1.8 \times 10^{-5} \frac{[1.01]}{[0.99]}$ and pH = 4.73 (note without buffer pH = 2!!!)

Buffers

- Buffers maintain an almost constant pH despite small amounts of strong acids or bases being added to the system.
- Buffers are usually made up of a weak acid and its salt or a weak base and its salt.
- Buffers play a critical roll in many biological and industrial processes
- E.g. hydrogen carbonate in blood

Molecular aspects of acid strength

	HOCl	HOBr	
Ka	2.9×10^{-8}	2.1×10^{-9}	for $HOX(aq) + H_2O(l) = XO^-(aq) + H_3O^+(aq)$

Molecular aspects of acid strength

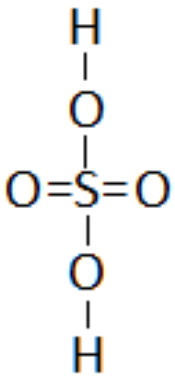
	HOCl	HOBr
Ka	2.9×10^{-8}	2.1×10^{-9}
EN	Cl = 3.0	Br = 2.8

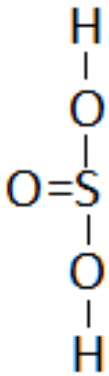
for $HOX(aq) + H_2O(l) = XO^-(aq) + H_3O^+(aq)$

Cl is more electronegative than Br so there is more of a partial positive charge on the H in HOCl making it easier for the H to come off as a Hydrogen ion.

The effect of a terminal oxygen

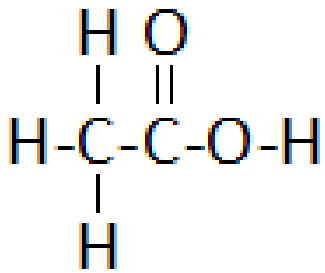
	H_2SO_4	H_2SO_3
Ka1	about 10^3	1.3×10^{-2}

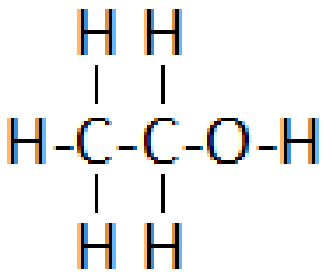




Resonance

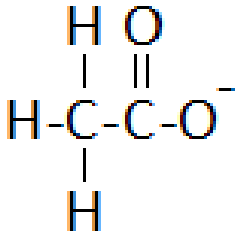
	CH_3COOH	CH_3CH_2OH
Ka	1.8×10^{-5}	1.3×10^{-16}

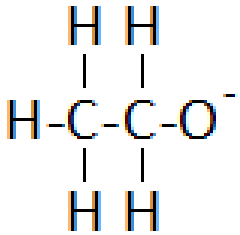




Resonance

	CH_3COOH	CH_3CH_2OH
Ka	1.8×10^{-5}	1.3×10^{-16}





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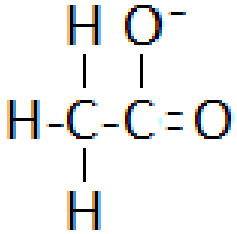

the negative

resonance